ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PRACTICE EXAM 3

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 1 style (theory, C++). Section A: objective questions (30) · Section B: problem solving with pseudocode and trace tables, data types, operators and expressions, selection and loops including switch, do-while, nested loops and input validation, functions, arrays, strings and classes (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. All program code must be written in C++ unless stated otherwise. Assume that #include <iostream>, #include <string> and using namespace std; are at the top of every program.
6. Write neatly and legibly. Indent your code so that it is easy to read.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The C++ operator that gives the remainder of a division is …

A / B % C # D &

1.1.2 The data type that stores a single character is …

A int B float C char D bool

1.1.3 The statement cout << 7 / 2; displays …

A 3.5 B 3 C 4 D 3.0

1.1.4 The loop that always runs at least once is the …

A for loop. B while loop. C do-while loop. D if statement.

1.1.5 The header file needed for cin and cout is …

A <iostream> B <cmath> C <string> D <vector>

1.1.6 After the declaration int marks[5]; the last element of the array has the index …

A 5 B 4 C 0 D 6

1.1.7 The keyword that leaves a loop immediately is …

A continue B return C exit D break

1.1.8 A variable declared inside a function is a …

A global variable. B constant. C local variable. D public member.

1.1.9 The member function with the same name as the class, which runs when an object is created, is the …

A destructor. B getter.
C main function. D constructor.

1.1.10 A single-line comment in C++ starts with …

A // B /* C # D --

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 C++ is case sensitive, so Total and total are two different variables.

1.2.2 A float can store whole numbers only.

1.2.3 The expression (5 > 3) && (2 > 4) is true.

1.2.4 A function declared as void does not return a value.

1.2.5 Array indexes in C++ start at 1.

1.2.6 The operator = assigns a value, while == compares two values.

1.2.7 A while loop tests its condition before each repetition.

1.2.8 Every C++ statement must end with a colon (:).

1.2.9 Private members of a class can be used directly in main().

1.2.10 A trace table helps to find logic errors by following the values of variables step by step.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Step-by-step set of instructions that solves a problem	A	string
1.3.2	Error that breaks the rules of the language and is found by the compiler	B	constant
1.3.3	Error that lets the program run but gives a wrong answer	C	algorithm
1.3.4	Named memory location whose value can change	D	switch
1.3.5	Named value that cannot change while the program runs	E	logic error
1.3.6	Value passed into a function	F	class
1.3.7	Statement that compares one variable with many fixed values	G	variable
1.3.8	Blueprint from which objects are created	H	compiler
1.3.9	Sequence of characters, such as "Robot"	I	syntax error
1.3.10	Translates the whole source code into machine code before it runs	J	parameter

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: PROBLEM SOLVING

2.1 Name the THREE basic control structures used in all programs. (3)

2.2 Write pseudocode for a program that inputs the readings of 10 temperature sensors, calculates the average and displays "OVERHEAT" if the average is more than 60. (6)

2.3 Draw a trace table for the code below that shows the values of i, a and b after each repetition, and write down the output. (5)

```
int a = 1, b = 0;
for (int i = 1; i <= 4; i++) {
  b = b + a;
  a = a * 2;
}
cout << a << " " << b;
```

[14]

QUESTION 3: DATA TYPES, OPERATORS AND EXPRESSIONS

3.1 Give the most suitable C++ data type for EACH of the following: (a) the number of wheels on a robot; (b) a battery voltage such as 7,4; (c) whether a motor is running or not; (d) the name of the robot. (4)

3.2 Given int x = 17; and int y = 5; write down the value of EACH expression: (a) x / y; (b) x % y; (c) (float) x / y; (d) x > 10 && y == 5; (e) x - y * 2. (5)

3.3 Explain the difference between = and == and give an example of each. (2)

3.4 Write C++ statements that declare a constant MAX_SPEED with the value 255, read an integer speed from the keyboard, and set speed equal to MAX_SPEED if the value entered is larger than MAX_SPEED. (5)

[16]

QUESTION 4: SELECTION AND LOOPS

4.1 Rewrite the following code using a switch statement. (6)

```
if (mode == 1) cout << "Forward";
else if (mode == 2) cout << "Reverse";
else if (mode == 3) cout << "Stop";
else cout << "Invalid";
```

4.2 Explain the difference between a while loop and a do-while loop, and give ONE situation where a do-while loop is the better choice. (4)

4.3 Write down the exact output of the following code. (3)

```
for (int r = 1; r <= 3; r++) {
  for (int c = 1; c <= r; c++) {
    cout << "*";
  }
```

```
  cout << endl;
}
```

4.4 Write a program segment that keeps asking the user to enter a distance until a value from 0 to 400 (inclusive) is entered, and then displays the valid distance. (5)

4.5 Explain what an infinite loop is and give ONE common cause. (2)

[20]

QUESTION 5: FUNCTIONS, ARRAYS, STRINGS AND CLASSES

5.1 Write a function cmToInch() that receives a length in centimetres as a float and returns the length in inches (1 inch = 2,54 cm). Then write the statement in main() that calls the function for 30 cm and displays the answer. (5)

5.2 The array float readings[5] = {12.5, 14.0, 13.2, 15.8, 11.9}; holds sensor readings. Write code that uses a loop to find and display the largest reading. (5)

5.3 Given string name = "RoboKit"; (a) what does name.length() return? (b) write the expression that gives the first character of name. (2)

5.4 Study the class below and answer the questions.

```
class Battery {
  private:
    float voltage;
  public:
    Battery(float v) { voltage = v; }
    bool isLow() { return voltage < 6.8; }
};
```

5.4.1 Name the private data member of the class. (1)

5.4.2 What is Battery(float v) called, and what does it do? (2)

5.4.3 Write code in main() that creates a Battery object with a voltage of 6,5 V and displays "Charge now" if the battery is low. (3)

5.4.4 Explain why voltage is declared as private. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER